

UPI & DIGITAL PAYMENT GROWTH IN INDIA

How has India's digital payment ecosystem grown, and what demographic and regional factors drive or hinder UPI adoption?

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GitHub: [mahmudalam0506-netizen/UPI-payment-Growth-analysis: Analysis of UPI growth trend and regional drivers in India\(2016-2026\)](https://github.com/mahmudalam0506-netizen/UPI-payment-Growth-analysis: Analysis of UPI growth trend and regional drivers in India(2016-2026))

Date: 10-07-2026

Summary

This project examines the growth of India's Unified Payments Interface (UPI) from 2016 to 2026 and investigates the demographic and regional factors that drive or hinder its adoption across states. Using a mixed-method approach, the analysis combines a time-series study of national UPI transaction trends with a cross-sectional correlational analysis of nine sampled states. National-level data shows UPI has grown from a niche payment method to India's dominant digital payment mode, with over 797.77 billion combined transactions recorded and bank participation expanding from 21 to 731 institutions. At the regional level, internet penetration ($r = 0.78$) and urbanization ($r = 0.71$) both show strong positive correlations with UPI adoption, with internet access emerging as the more consistent enabler. The analysis is descriptive and correlational rather than causal, and the regional sample — drawn from 9 purposively selected states — may overstate the true strength of these relationships nationally. The project was built using Python for statistical analysis, Power BI for an interactive dashboard, and Excel for data cleaning and integration.

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1. Introduction

1.1 Background

With Unified Payments Interface (UPI) emerging as the dominant mode of retail digital transaction, digital Payments have grown rapidly over the past years. It was launched by National Payments Corporation of India (NPCI) in 2016 and has enabled interoperable bank-to-bank transfers through smart phones. Since its launch, UPI has risen from small payment method to backbone of nation's digital payment method processing hundreds of billions of transactions annually. This growth raises question on factors affecting the growth and whether adoption is uniform across the country.

1.2 Problem Statement

UPI's national level growth being well documented, less information is available about demographical and regional factors that assist or hinder adoption at state level. This analysis examines national trend in UPI growth over time as well as regional factors – especially internet penetration and urbanization – that affects the adoption rate across states.

1.3 Objectives

- Analyse the national trend in UPI transaction volume, value, and bank participation from 2016 to 2026
- Identify how average transaction value has evolved, reflecting shifts in usage patterns
- Examine the relationship between regional factors (internet penetration, urbanization) and UPI adoption across a sample of Indian states
- Present findings through statistical analysis (Python) and an interactive dashboard (Power BI)

1.4 Scope

Covered:

- National-level UPI data (2016–2026, including partial-year 2026 data)
- A 9-state regional sample correlating UPI per-capita adoption with internet penetration and urbanization

Not covered:

- Full 28-state analysis (official pre-2025 state-wise UPI data isn't published by NPCI)
- Causal analysis (only correlational relationships are examined)
- International UPI adoption

1.5 Report Structure

This report is organized as follows: Section 2 presents the data sources used. Section 3 explains the methodology used for cleaning, merging, and analysing the data. Section 4 presents findings, split into national trend analysis and regional adoption drivers. Section 5 covers the Power BI dashboard. Section 6 discusses limitations, followed by conclusions and recommendations in Section 7.

2. Data Sources

This project draws on the following data sources:

- **NPCI year-wise UPI statistics (2016–2026)** — national monthly transaction volume, value, and bank participation data, published by the National Payments Corporation of India
- **TRAI Quarterly Performance Indicator Report (QE June 2025)** — state-wise internet penetration data (rural, urban, and total), sourced from traigov.in
- **Dataful/Factly article citing NPCI-sourced data** — state-wise UPI per-capita transaction figures for FY 2024–25, covering a 9-state sample
- **Census 2011** — state-wise urban population percentages, used as the urbanization indicator in regional analysis

3. Methodology

3.1 Research design

This Analysis adopts a mixed-method approach, combining a Time-series analysis of national UPI transaction data with a cross-sectional correlational analysis of regional adoption drivers. The former examines how UPI usage has evolved between April 2016 and June 2026 while latter investigates how internet penetration and urbanization has affected the former. The regional analysis is exploratory and descriptive in nature rather than confirmatory.

3.2 Data Sources

- **National Payments Corporation of India (NPCI):** Retrieved month wise data from April 2016 to June 2026 for Banks Participation in UPI, Volume of UPI transactions (in millions), Value of UPI transactions (in crores).
- **Telecom Regulatory Authority of India (TRAI):** Accessed State-wise internet subscriber data (From Quarterly Performance Indicator Report, Quarter Ending in June 2025), For Internet subscriber per 100 population for Urban, Rural and total Populations.
- **NPCI (via Dataful/Factly analysis):** Accessed State-wise UPI per capita data (FY 2024-2025). Since NPCI's official state-wise reporting began only in June 2025, no historical state wise series is available, this analysis therefore uses most recent full year series published.
- **Census of India (2011):** Derived State-wise rural and urban percentage used as a proxy for regional analysis, this remains the most recent full Census available, as the 2021 Census has not yet been conducted.

3.3 Regional Sample Selection

Due to absence of freely available state-wise UPI dataset consisting of all 28 states, an alternative “leaders versus laggards” sampling strategy was used. In this method, 9 states comprising of 5 adoption leading states (Telangana, Delhi, Goa, Karnataka, Maharashtra) were contrasted with 4 lower adoption states (Uttar Pradesh, Bihar, Manipur, Tripura) to identify adoption drivers. Because data sample was carefully chosen for its extreme values rather than being chosen randomly, the correlation coefficient observed during study might overstate the strength of association observed during full or random data sample. This is inherent limitation of the sampling strategy, not the underlying relationship.

3.4 Data Integration

State-wise UPI adoption figures, TRAI internet penetration data, and Census urbanization percentages were merged using state name as the common key. Because these three datasets belonged to different Time periods — TRAI (June 2025), UPI adoption (FY 2024-25), and Census (2011) — the merged dataset reflects an approximate rather than a precisely synchronized snapshot in time. This adjustment is acceptable given that internet penetration and urbanization are slow-changing variables, but is noted here as a methodological constraint. Additionally, approximately 26–27% of national UPI transactions are classified by NPCI as "unclassified" (state-unattributable) and were excluded from all per-state calculations, consistent with standard treatment of this dataset in existing research.

3.5 Data Cleaning and Validation

Raw NPCI dataset was collected in separate yearly files in inconsistent format. Raw Dataset files were cleaned and formatted (text to number format, standardized date format, and remove non-printing characters) in Microsoft excel separately. Clean data was validated by ensuring no empty value or duplicate value was present in any of the datasets. Cleaned and validated yearly files were later combined into single master file using Power Query. In master dataset, a separated Source. Name column was maintained to track the originating file of the row data.

3.6 Analytical Techniques

- **Trend Analysis:** Year-over-year (YOY) growth of value as well as volume of UPI transactions have been calculated as percentage growth as from the prior year. In Year-over-year growth graph, initial years (2016 and 2017) has been removed due to extremely low base values, which caused disproportionate growth percentage that obscured meaningful patterns in further years. This inappropriate pattern is due to early adoption artifact rather than a meaningful trend.

- **Average transaction value:** Calculated as total transaction value divided by total transaction volume for each time period. This creates a volume weighted average rather than an individual transaction value weighted average, meaning period with more transaction volume contribute more to results.
- **Correlation analysis:** The Pearson correlation coefficient was used to measure the strength and direction of the linear relationship between (a) internet penetration and UPI adoption, and (b) urbanization and UPI adoption, across the 9 sampled states. Given the sample size was small, these coefficients are reported descriptive and have not been tested for statistical significance; they should be interpreted as indicative rather than confirmatory. Because internet penetration and urbanization are themselves likely correlated, these two relationships were assessed independently (bivariate) rather than jointly, and this study does not isolate the independent effect of either variable while controlling for the other.
- **Geographic Visualization:** State-wise adoption was visualized using interactive choropleth map (Python, Folium library) and a filled map visual (Power BI).

3.7 Scope

The national trend analysis covers April 2016 to June 2026. Noting that 2026 reflects only six months of data (as this is the time of analysis and preparation of report) and is treated as a partial year throughout this study to avoid misleading year-over-year comparisons.

3.8 Analytical Tools

- **Microsoft Excel:** data cleaning, Power Query transformation, and formula-based merging (VLOOKUP)
- **Python (pandas, seaborn, matplotlib, folium):** trend analysis, correlation calculation, and geographic visualization
- **Microsoft Power BI:** interactive dashboard development for presenting findings

4. Data Analysis & Findings

4.1 National Trend Analysis

4.1.1 Transaction Volume and Value Growth (see Figure 1, Figure 2)

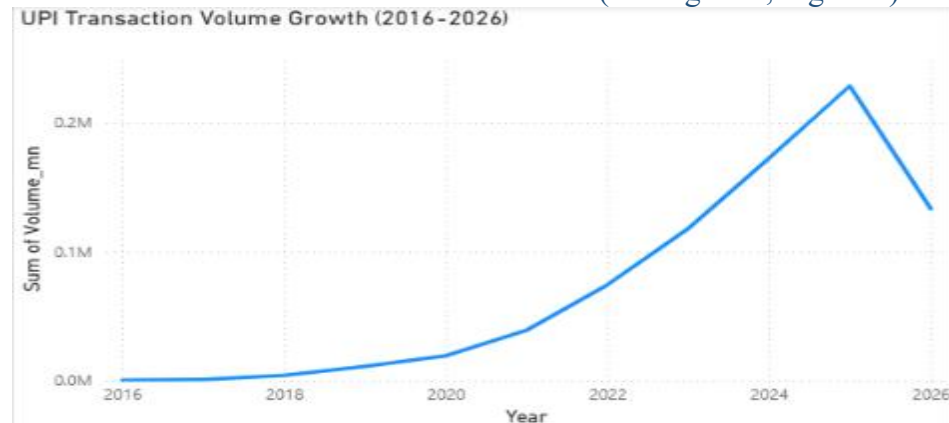


Figure 1

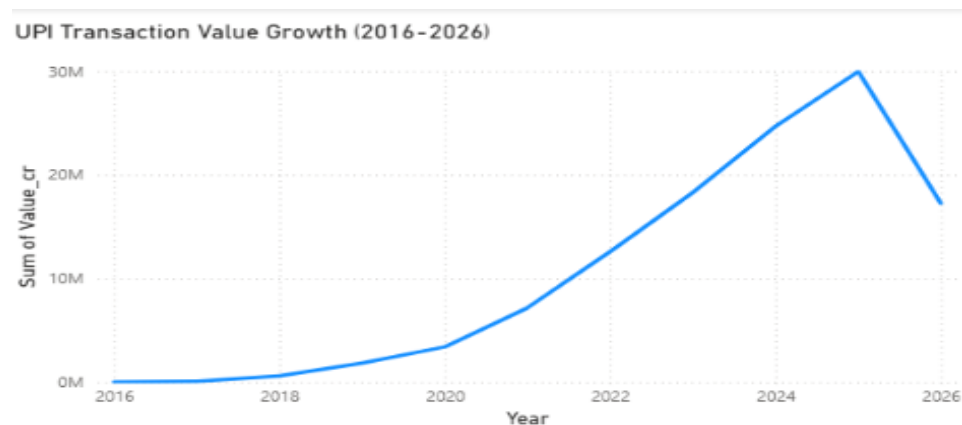


Figure 2

UPI transaction volume and value had grown considerably since platform's 2016 launch. Collectively, over 797.77 billion transactions have been recorded from April 2016 to June 2026. In 2026 alone, despite being a partial year transaction volume has reached 133 billion, highlighting UPI's adoption as habitual payment mode. Highest rate of transaction was observed in 2018 and remained strong till 2024 showing UPI's transition as emerging payment method to India's dominant mode of digital transaction. These figures were independently derived in both the Python analysis and the Power BI dashboard, with consistent results across both.

Note that data for year 2026 is available only till June month, the decline in volume and value of transaction is a partial artifact rather than being an actual fall down.

4.1.2 Bank Participation (see Figure 3)

Bank Participation Growth (2016-2026)

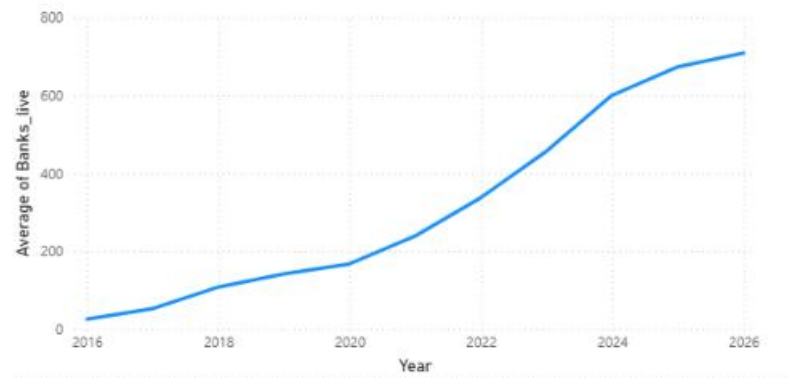


Figure 3

The number of banks live on UPI platforms have grown exponentially from 21 in 2016 to 731 in 2026. Growth of banks participation occurred in three distinct phases: slow, steady growth from 2016 to 2022 (approximately 20 to 250 banks); sharp acceleration from 2022 to 2024 (250 to approximately 600 banks); and a plateau from 2024 onward (650–730 banks). This suggests maximum eligible banking institutions have approached UPI innovation.

4.1.3 Average Transaction Value (see Figure 4)

Average UPI Transaction Value Over Time (₹)

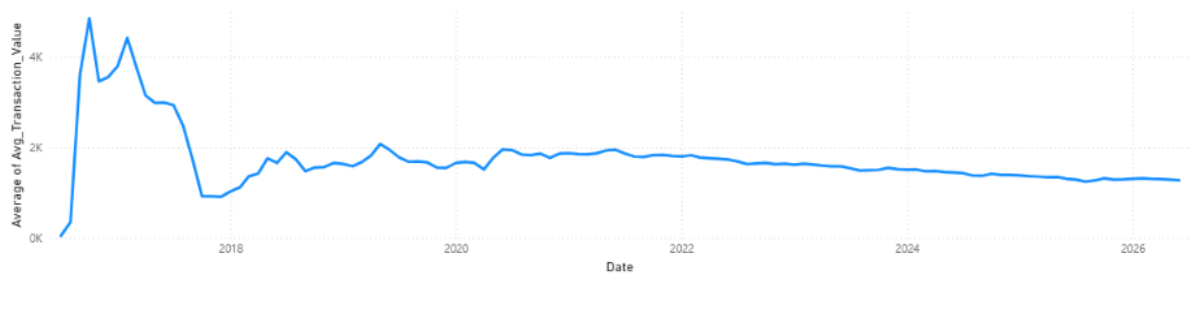
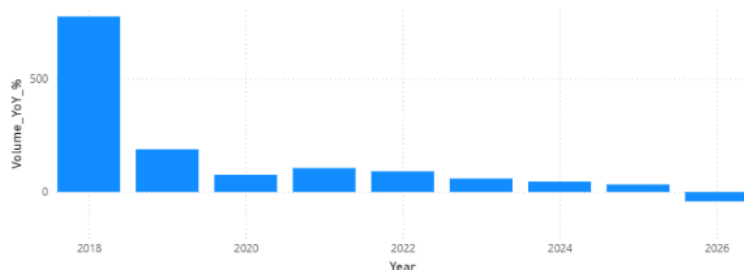


Figure 4

The average value per UPI transaction has declined from approximately ₹1,500 in 2018 to under ₹1,000 by 2025–26 (Year 2016 outliers average transaction value chart due to lower transaction volume). This decline of average value suggests UPI's adoption is being shifted from a payment method for large, occasional transactions to consistent everyday small transactions such as Grocery shopping and transport expenses.

4.1.4 Year-over-Year Growth (see Figure 5)

UPI Volume Year-over-Year Growth (%) — 2018 Onward



NOTE:-

- 2026 reflects data through June only (partial year); growth % may understate full-year performance.
- Initial years are excluded due to near zero - base year distortion.

Figure 5

Year-over-year transaction growth peaked in 2018 (772.96%) and has tapered over subsequent year: 187.95% (2019), 75.02% (2020), 105.21% (2021), 91.11% (2022), 58.88% (2023), and 46.38% (2024), reaching 32.56% in 2025. This consistent tapering suggests technology thriving from early adoption phase to mainstream, widespread use. Negative figure in 2026 (-41.74%) is due to availability of only six months of data against a full prior year and should not be considered as an actual decline. Data from 2016-2017 was excluded from visualization due to extremely low base-year values producing statistically extreme, non-representable percentage changes (exceeding 16,000% in 2017).

4.2 Regional Adoption Drivers

4.2.1 Internet Penetration and UPI Adoption (see Figure 6)

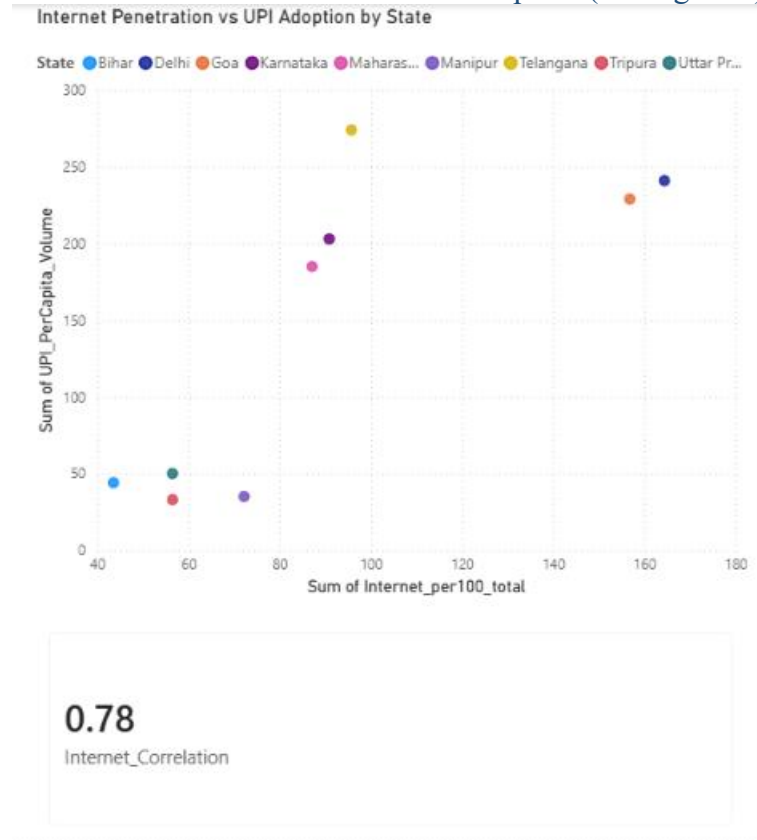


Figure 6

Across the 9 states sampled in this study, internet penetration ranged from 43.55 (Bihar) to 164.42 (Delhi) subscriptions per 100 population, while UPI per-capita transaction volume ranged from 33 (Tripura) to 274 (Telangana) transactions per person annually. These two observations showed a strong positive relationship (Pearson $r = 0.78$). States with higher internet subscription density (approximately 87 per 100 population) have consistently shown higher UPI adoption, while states with lower internet subscription density (approximately 73 per 100 population) showed lower adoption. As this figure is derived from sample-selected database and thus, it may overstate the actual strength between internet penetration and UPI adoption.

4.2.2 Urbanization and UPI Adoption (see Figure 7)

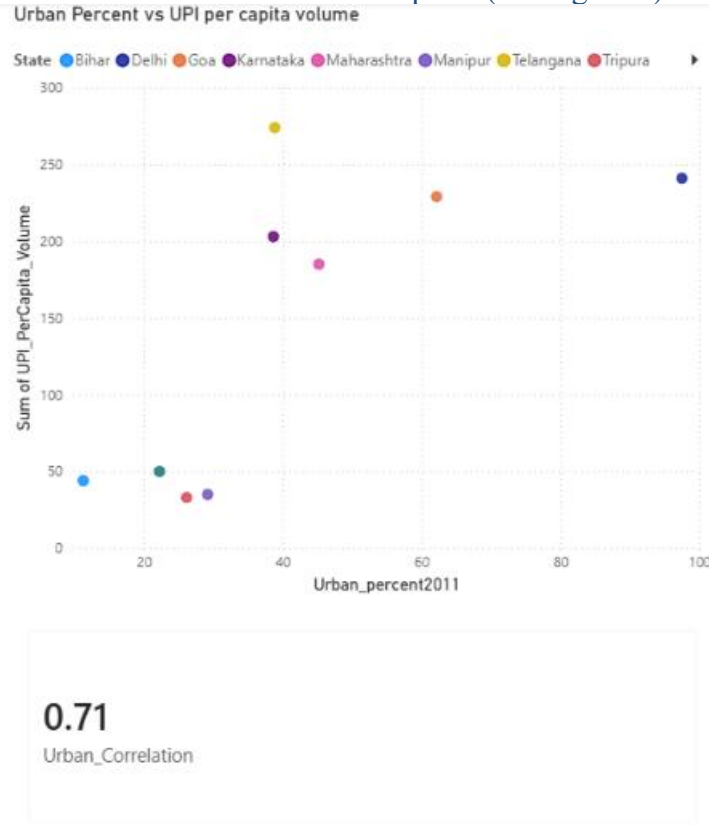


Figure 7

Urban population percentage (Census 2011) has shown a strong positive relationship with UPI adoption (Pearson $r = 0.71$), although this relation is slightly weaker than internet penetration. Notably, Telangana having highest UPI adoption in the sample has moderate urbanization (38.88%), which is much lower than states that have lower adoption with higher urbanization, including Delhi (97.5%) and Goa (62.17%). This suggests internet access is a strong and consistent enabler for UPI adoption rather than urbanization itself. As stated in internet penetration correlation, this figure is also a result of sample selected datasets and thus, it may overstate the actual relation between UPI adoption and Urbanization.

4.2.3 Geographic Patterns (see Figure 8)

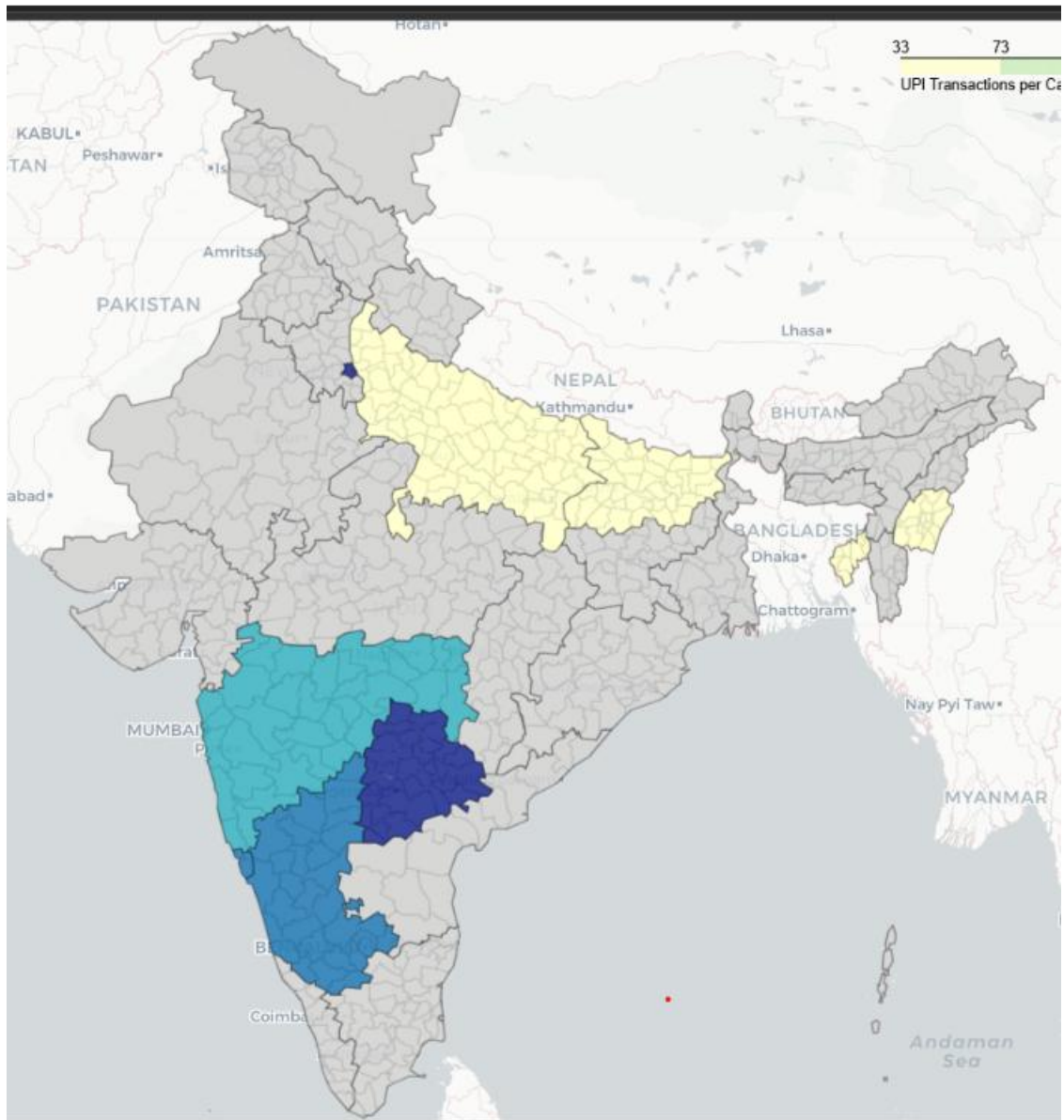


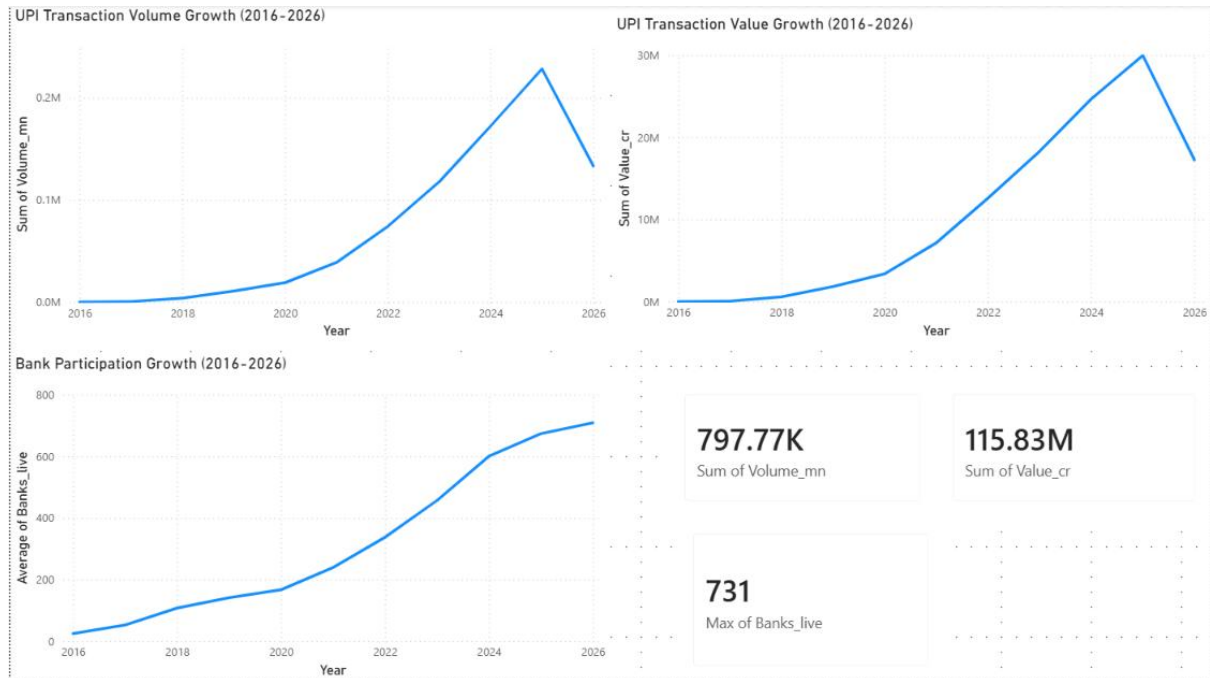
Figure 8

Geographical visualization of adoption revealed a regional clustering effect: high adoption is observed in southern and western India (Telangana, Karnataka, Maharashtra, Goa) and the National Capital Region (Delhi), while low adoption is observed in parts of northern India (Uttar Pradesh, Bihar) and the northeast (Manipur, Tripura). States other than 9 sample states are shown without shading on map, stating the absence of comparable per-capita UPI data for those regions rather than an assumption of zero or low adoption.

5. Dashboard Overview

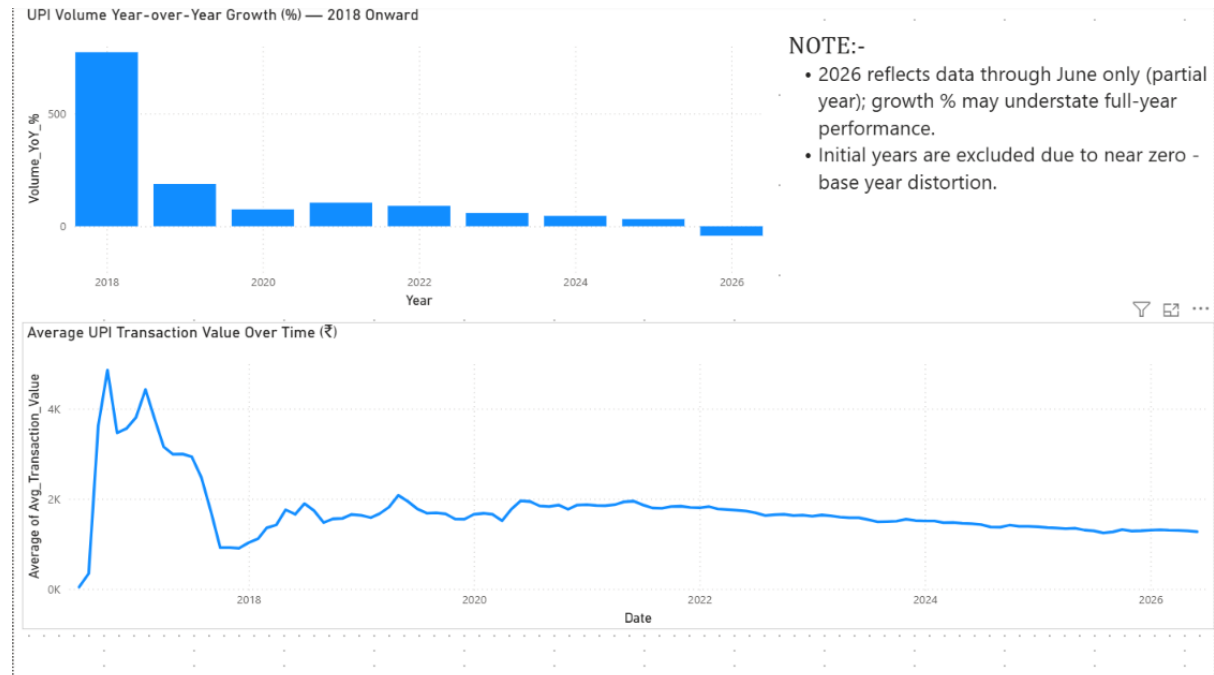
To appreciate the statistical analysis, an interactive Power BI dashboard was made to allow study of UPI adoption trends and regional patterns. The dashboard consists of four pages:

5.1 National Trend Page



This page contains visualizations of UPI's national growth from 2016 to 2026, including transaction volume, transaction value, and bank participation over time. It also includes sum of total transaction volume, sum of total transaction value, and total bank participating in UPI over the time period of 2016 to 2026 displayed in separate section. Time-intelligence measure was built in Power BI using a separate calendar table to support year-over-year comparisons. Bank participation is measured as average or max aggregation rather than sum as it represents a point in time count.

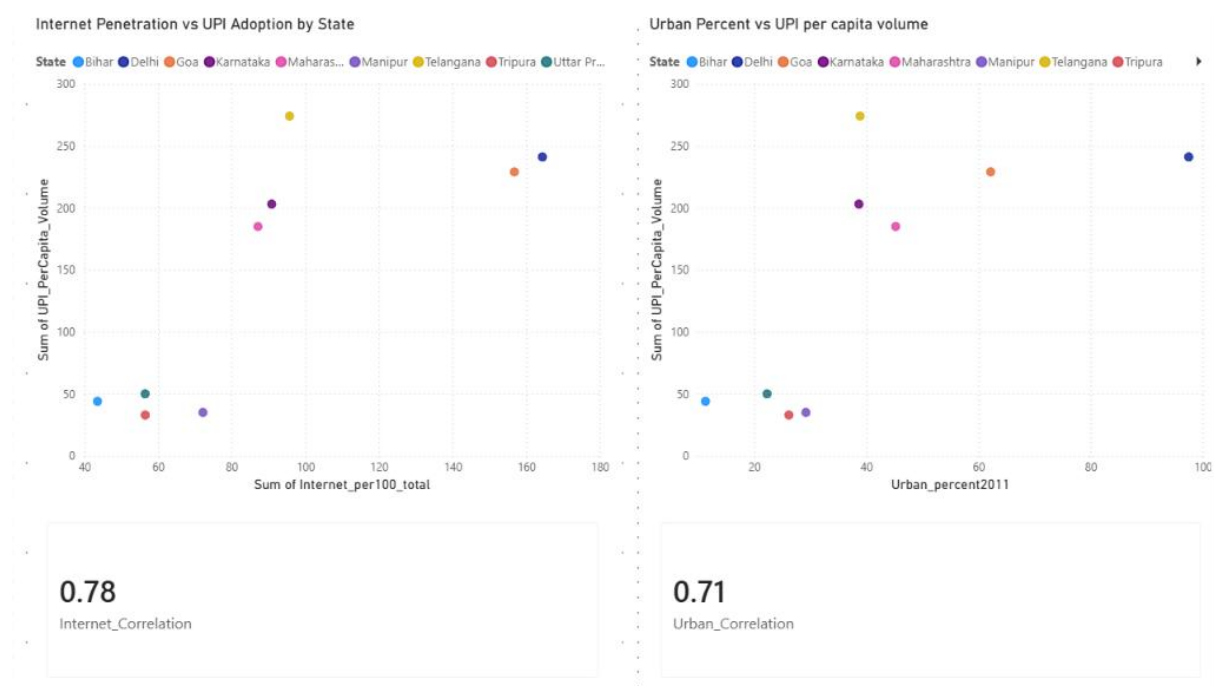
5.2 Transaction Behaviour Page



This page visualizes year-over-year growth of UPI's transaction volume as well as UPI's average transaction value over years. Decline in UPI's average transaction value reflects UPI's shift from occasional, higher value transaction in early years to smaller, frequent transaction in recent years. Again, time-intelligence measure was used here to compare year-over-year growth.

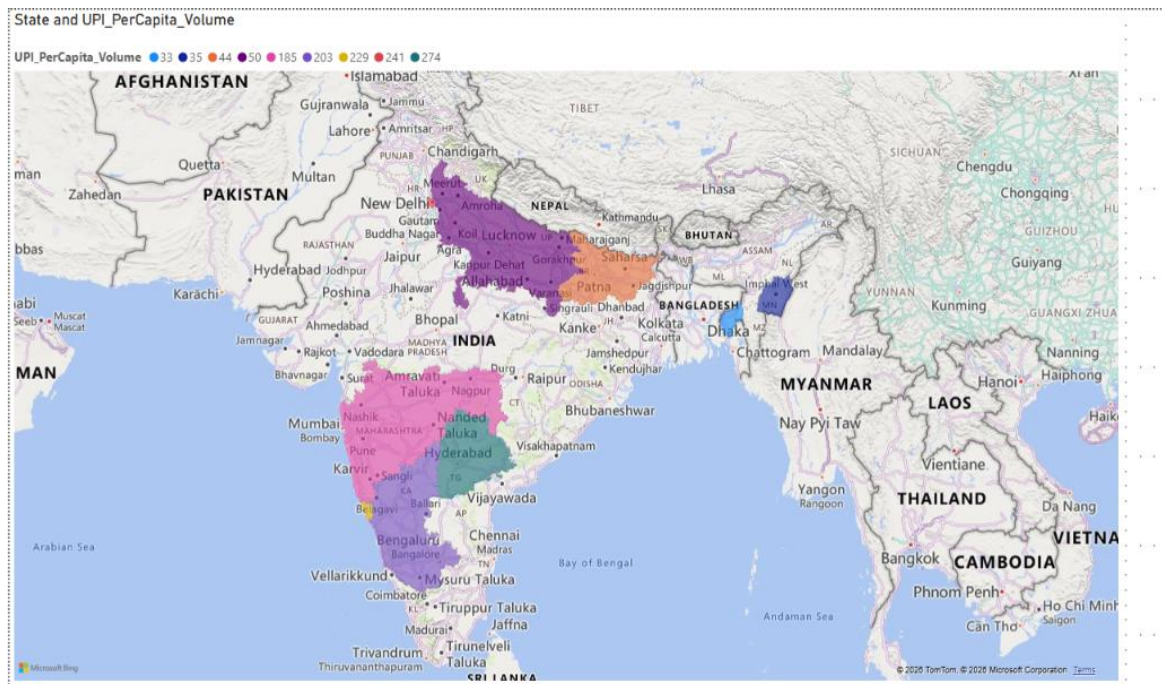
Note: Initial years are excluded due to near zero or extremely low base years which were distorting the later years graphs representation. 2026 is a partial year and may understate the full year growth.

5.3 Regional Drivers Page



This page contains visualization of regional factors (internet penetration and urbanization) correlation with UPI's adoption for the 9 sampled states using scatter plot chart, allowing comparison between high and low adoption states. The correlation coefficient was calculated within the model and may overstate the correlation over the full 28 states observation.

5.4 Map View Page



This page contains filled (choropleth) map of India showing state-wise UPI adoption, providing a geographic overview of nation's UPI adoption pattern alongside the UPI per capita transaction volume for each state.

6. Limitations

- **Sampling limitation:** The regional analysis used a selected 9-state “leaders versus laggards” sample rather than random or complete 28 states data (due to absence of free dataset). This might lead to observed correlation coefficients (internet penetration: $r = 0.78$, urbanization: $r = 0.71$) overstating the relationship’s original strength across whole of India.
- **No statistical significance testing:** Correlation coefficients are reported as descriptive only. Due to small sample size, relations were not tested for statistical significance and should be read as indicative rather than confirmatory.
- **Variables tested independently:** The variables, internet penetration and urbanization were analysed separately (bivariate) instead of jointly, since these two are likely correlated with each other. So, the effect of one cannot be stated when other is controlled and vice versa.
- **Mismatched time periods across datasets:** The three datasets used in analysis comes from different time periods — TRAI internet data (June 2025), UPI per-capita data (FY 2024–25), and Census urbanization data (2011). This is due to delay of latest 2021 census survey and absence of pre 2024 per-capita UPI data.
- **Outdated urbanization baseline:** Census 2011 being the latest full Census available (2021 Census has not been conducted), meaning the urbanization figures used are over a decade old relative to the 2024–26 UPI data they are compared against.
- **Unclassified transactions excluded:** NPCI has marked approximately 26-27% of data "unclassified" (state-unattributable) and so, these transactions were not involved in per-state calculations.
- **Partial-year 2026 data:** Data for year 2026 was unavailable post June, so all 2026 figures or values are observations of partial year rather than a full year observation.
- **No causal claims:** All relationships reported in this project (e.g., internet penetration and UPI adoption) are correlational, not causal. No claim is made that one factor directly causes another.

7. Conclusions & Recommendations

7.1 Conclusions

- UPI has grown from an occasional payment method in 2016 to India's dominant digital payment method, with over 797.77 billion total transactions recorded from April 2016 through June 2026.
- UPI's growth has decelerated over time: year-over-year volume growth peaked at 772.96% in 2018 and has fallen to around 32–33% by 2025, indicating a shift from early explosive adoption to a more mainstream, steady-growth phase.
- Participation of banks has grown over time from 21 banks in 2016 to over 730 by 2026. Bank participation growth occurred in 3 phases: slow growth phase (2016–2022), steep acceleration phase (2022–2024), and a plateau from 2024 onward, suggesting most eligible banks have now joined the UPI ecosystem
- Average transaction value has declined from roughly ₹1,500 (2018) to under ₹1,000 (2025–26), showing UPI has shifted from occasional large transfers toward everyday small-value payments like groceries and transport.
- At the regional level, internet penetration showed a stronger and more consistent relationship with UPI adoption than urbanization did. Telangana, for example, has the highest UPI adoption in the sample despite only moderate urbanization (38.88%) — suggesting internet access may be a more direct enabler of adoption than urbanization itself.

7.2 Recommendations for Future Work

- Repeating regional analysis when complete state-wise UPI data for 28 states becomes available from NPCI, to examine whether the observed correlation is applicable at national scale.
- Using updated urbanization data (after the availability of next census) to minimize the time-mismatch limitation between UPI and urbanization figures.
- Using multivariate regression analysis to independently study the effect of one variable by keeping the others constant.
- Extending the analysis by including more variables (e.g., income levels, smartphone penetration, age distribution) that too can affect the UPI's regional adoption.

References

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